Unit 3: Macroeconomics

Overview

September 27, 2025

Micro- and macro-economics

- In microeconomics, we studied behavior of individual agents of economy
- In macroeconomics, we shall study behavior of the economy as a whole
- Specifically, the macro-aggregates, their relationships and determinants
  - Business cycles: short-term fluctuations in the macro-aggregates
  - Economic growth: long-run changes in output and living standards
- Although, to understand their movements, the macro-aggregates often need to be disaggregated
  - Public and private sectors
  - Agriculture, Industry and Services
  - Infrastructure vs others
  - Prices of food and non-food items
  - Labor market conditions in formal and informal sectors

The birth of macroeconomics (1)

- 1930s: the great depression and Keynes' *General Theory*
  - In the 1930s, production, employment, and prices collapsed in the US and many industrial countries
  - Keynes developed a theory that helped explain the factors behind the fluctuations and suggested what governments can do to control such situations
  - From a Keynesian perspective, offsetting disturbances (that originate in the private sector) to the aggregate demand is a major stabilizing role of the government
- 1945: World War II [production facilities: civilian -> military]
- 1960s: Vietnam War [slow growth and inflation]
- 1970s: Oil Crisis [again, slow growth and inflation]
- 1990s: rapid growth ['fabulous decade']
- 2000: dot-com bubble [sharp decline in the prices of technology stocks, following substantial investments and a sharp rise in their price]
- 2008-10: US Financial Crisis [subprime lending; asset prices]

The birth of macroeconomics (2)

- Macroeconomic issues have dominated the political and economic agenda for a large part of the 20th Century in the US
- Macroeconomic failures have resulted in political turmoil
  - Leaders from the erstwhile Soviet Union proclaimed that they would overtake the West economically
  - Macroeconomic failures led to disintegration, despite Soviet Union having a strong military and possessing natural resources
  - Many economies in the Eastern Europe met with a similar fate
  - These events helped emphasize the role of markets (vis-à-vis state)

Keynes

- Academia, Administration and Governance, and Policy
  - Professor at Cambridge; also served the UK Treasury
  - Institutions: Royal Economic Society (Secretary) and Editor, *EJ*
- Academic contributions include,
  - Indian Currency and Finance (1913)
  - Treatise on Money (2 vols, 1930)
  - The General Theory (1936)
- Keynes and India
  - After clearing the (British) Civil Service, he was appointed in 1906 in the India Office at the London; got familiar with the problems of Indian currency during First World War (Keynes never visited India)
  - Member, Royal Commission on Indian Finance and Currency
  - The M & MP reports (The Material and Moral Progress and Condition of India): these were the annual official reports of the India Office to the British Parliament; published for several years during 1860s to 1930s; Keynes edited the report for 1906-07 and criticized the appointment of certain officials made in India

- Source: *The Collected Writings of J M Keynes*, Vol XV, Cambridge University Press, 2013

The General Theory

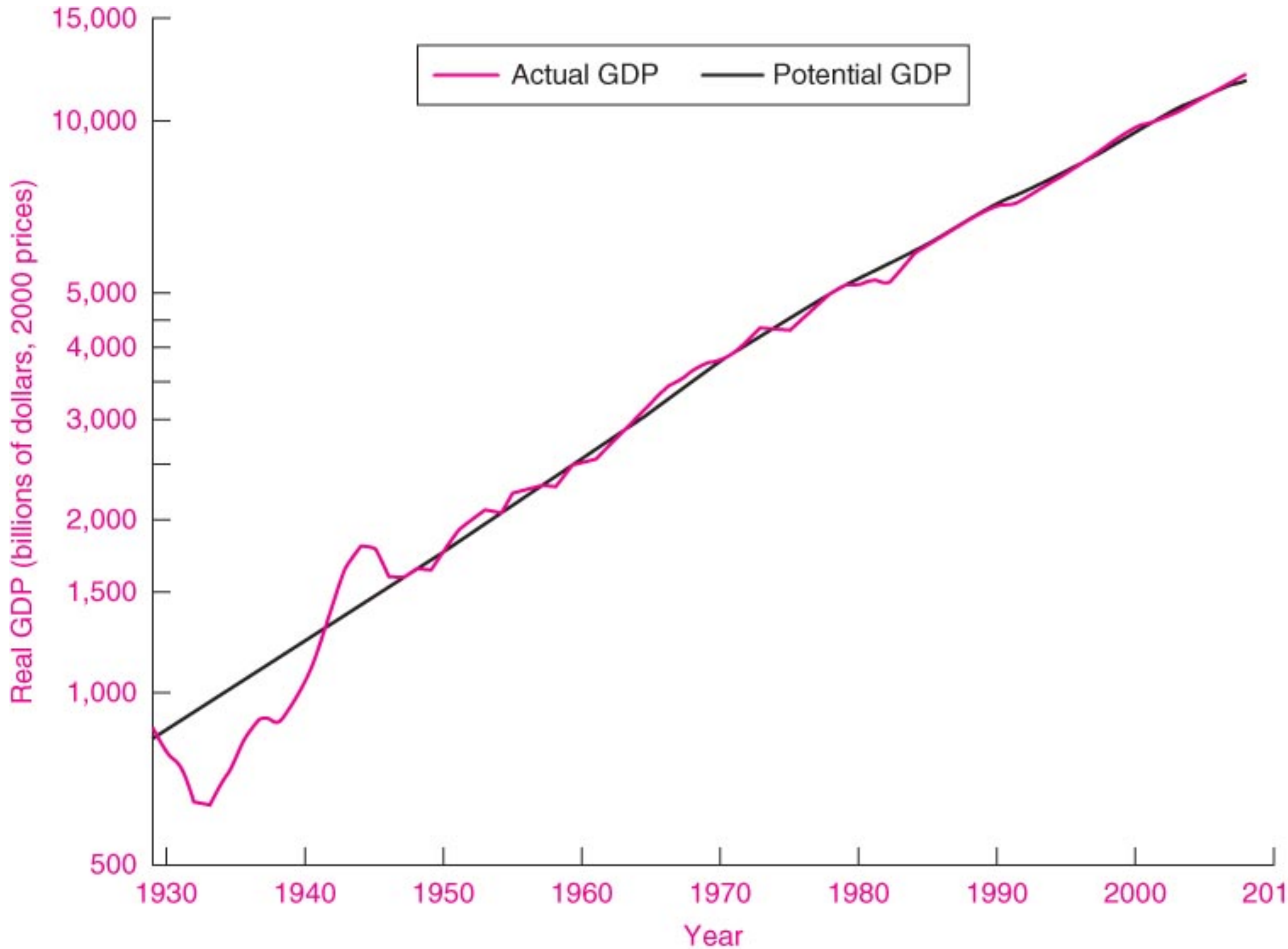
- What is “General” about it?
  - Classicals – Ricardo, Mills, Marshall, Pigou
  - Prior to Keynes, it was believed that macroeconomy can work on its own. The government need not intervene and leave the economy alone (the “invisible hand” guides an economy)
  - Keynes argued that the classical postulates “are applicable only to a special case only and not to the general case”… “Moreover, the characteristics of the special case assumed by the classical theory happen not to be those of the economic society in which we actually live”
  - Classicals assume full-employment and wage-price flexibility
- The General Theory
  - Employment: The demand and supply are functions of employment, their intersection
  - Interest: Saving and investment
  - Money: the demand for money; liquidity
- Keynesian views have had a strong influence on the macroeconomic policy across the globe
  - The governments recognized the importance of economic growth to improve the standard of living
  - The US government enacted the Employment Act, 1946
  - India: the Task Force on Minimum Needs and Effective Consumption Demand (1979) sought to understand the consumption patterns through improved methodology and database
- Source: *The Collected Writings of J M Keynes*, Vol VII, Cambridge University Press, 2013

US real GDP growth rate, 1929-2008

Goals of macroeconomics (1)

- How do we evaluate success of an economy?
- Output
  - The ultimate aim of economic activity is to provide the goods and services that the population desires
  - We need a summary measure of production of various goods and services in an economy
  - Economic activity is measured using gross domestic product (GDP) which is the value of all final goods and services produced in a country during a given period
  - GDP and economic growth (more in the next class)
  - Potential GDP is the maximum level of output an economy can produce, given its endowments, while maintaining price stability
  - Actual GDP is subject to business cycles but the potential GDP is not

Actual and potential GDP; business cycles occur when actual output departs from its potential level



Goals of macroeconomics (2)

- Employment
  - In an economy, it is desirable to have high employment or low *un*employment
  - The unemployment rate is defined as the share of unemployed in the *labor force*
  - Labor force excludes those who are not looking for a job
  - High unemployment also has welfare consequences
- Prices
  - The third goal of macroeconomics is price stability
  - Inflation is the percentage increase in the general price level from one year to the next

Hyperinflation: using wheelbarrows to carry money for shopping!

Goals of macroeconomics (3)

- Prices
  - Hyperinflation is a rise in price level which is of very high magnitude
    - Germany in 1920s: following the First World War, had to borrow heavily to pay for the costs of War
    - Brazil in 1980s: food price shocks and (external) public debt
    - Presently, Zimbabwe: Political instability; land reforms (beneficiaries were not trained farmers)
  - Rate of inflation can be negative and such an episode is known as deflation
  - It is desirable to have low and stable inflation
- Macroeconomic stability is desirable not only for its own sake but also for the long run growth of an economy
- What about education, health, poverty, etc?
  - These social goals are not the primary concerns of the macro policy. The macro stability and higher growth do facilitate achievement of these goals

Goals: summary

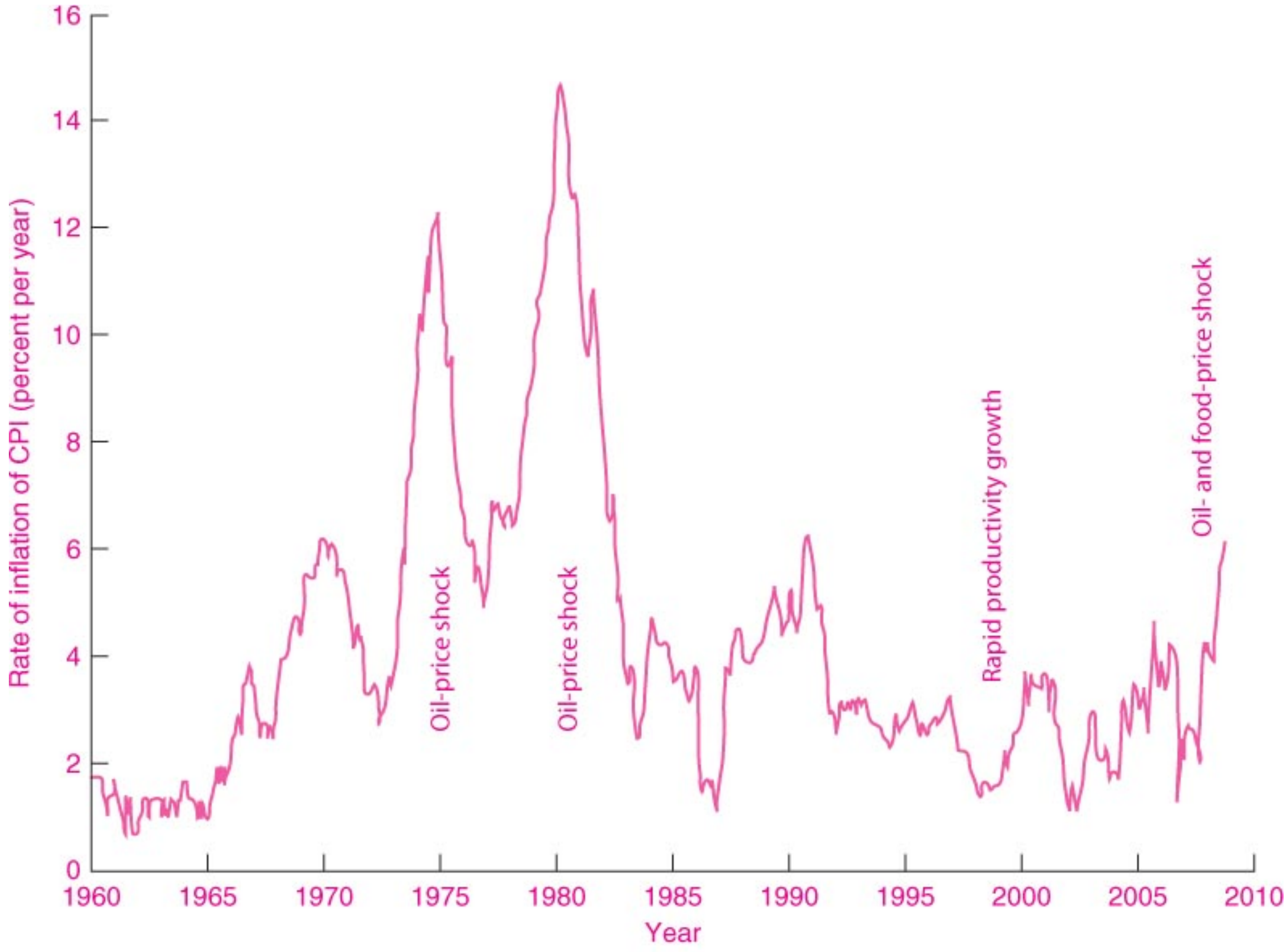
- Output
  - High level of output
  - Higher growth of output
- Employment
  - Low unemployment
- Price level
  - Price stability
- The above three often move in tandem
  - 1930s: unemployment was almost one-fourth of the workforce in the US and industrial production fell by one-half
  - 1990s: Japan – the lost decade; strong post-war growth led to asset price bubble. Inflation started rising, which led the central bank to initiate some measures. These, in turn, busted the bubble. The problem of bad loans, issues with financial market, ultimately resulted in economic stagnation and high unemployment
  - 2020: Covid affected nearly all major economies of the world

Recession and depression

- Recession
  - A period of significant decline in output (decline in two successive quarters)
- Depression
  - A prolonged period characterized by low output and high unemployment
  - Generally, recession lasting for a period more than two years

Unemployment and recessions

Inflation; US CPI



Instruments of macroeconomics (1)

- Instruments: the policy tools available with the government to affect the goals
- **Fiscal policy**
  - Government budget
  - Use of government expenditure and taxes
  - *Government expenditure*: government purchases (spending on goods and services; e.g., public infrastructure, salaries), and transfer payments (e.g., interventions aimed at targeted groups)
  - Government expenditure affects the overall level of spending in an economy and thus the level of GDP
  - Government spending determines the relative size of the public and private sectors
  - Importance of public expenditure: unlike private expenditure, public expenditure usually benefits society

Instruments of macroeconomics (2)

- **Fiscal policy (contd.)**
  - *Taxes*: affect the disposable income of the people which in turn determines consumption and saving
  - Taxes also affect the prices of goods and services and thus affect incentives and behavior
  - Fiscal policy is typically used to affect economic growth through its impact on savings and investment
  - Nature of taxes and expenditure have significant macroeconomic implications. So also, is the balance between expenditure and revenue (which could lead to a debt crisis)
- **Monetary policy**
  - Operates through money, credit, and banking system
  - Banks: what if they increase the interest rates on deposits
  - RBI: regulates the money supply in the Indian economy; sells and buys government securities

Instruments of macroeconomics (3)

- **Monetary policy**
  - Generally when inflation is perceived to be to high, the RBI raises interest rates. This leads to reduction in consumption and investment and lowers GDP and inflation
  - But, reduced consumption and investment means lower growth (GDP). Hence, there seems to be a trade off between inflation and economic growth
  - Open economy: exchange rates control; this role has become important with increasing trade and financial linkages
  - More often, monetary policy is used to stabilize the business cycles

Aggregate Demand (AD)

- AD:
  - It is the total amount that different sectors of the economy are willing to spend in a period and equals total spending on goods and services
  - Depends on prices, policy, and other factors
- Components
  - Consumption
  - Investment
  - Government expenditure
  - Net exports

Aggregate Supply (AS)

- AS
  - It is the total quantity of goods and services that the producers are willing to supply in a country in a period
- Depends on
  - Prices and costs
  - Productive capacity of the economy
- The intersection of the AD and AS gives the equilibrium
  - National output and overall price level
  - Also, the employment

The equilibrium

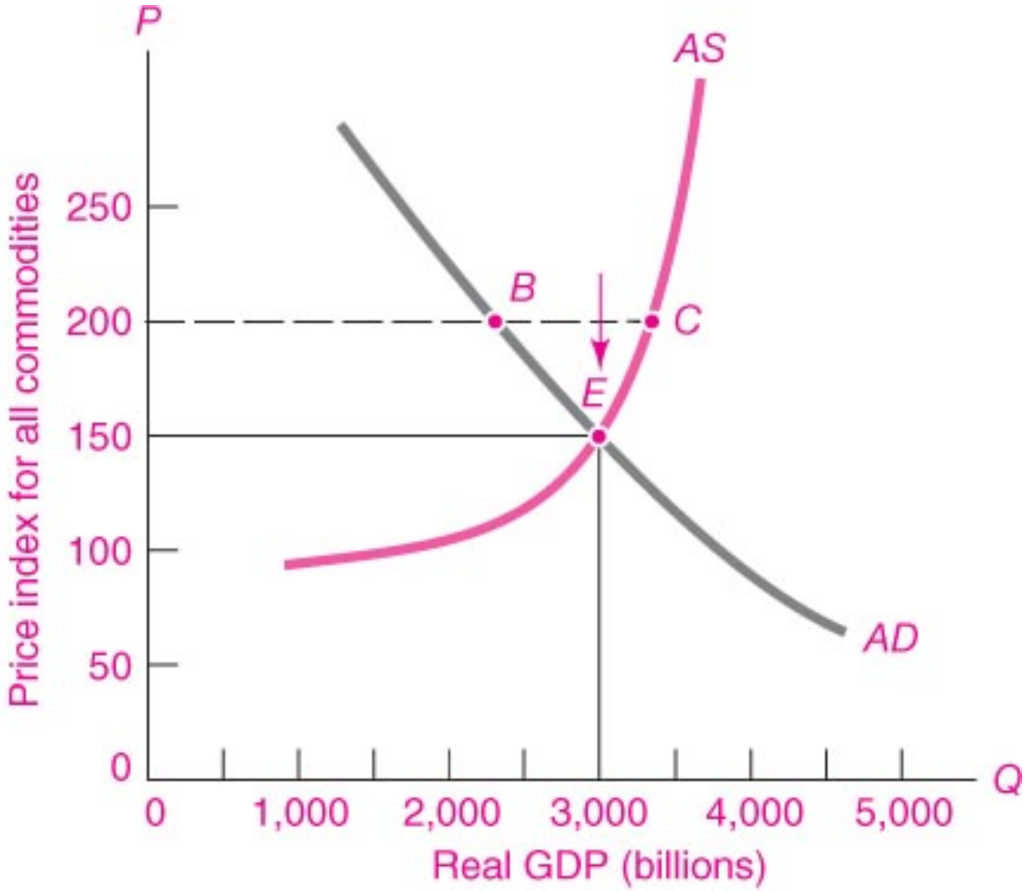


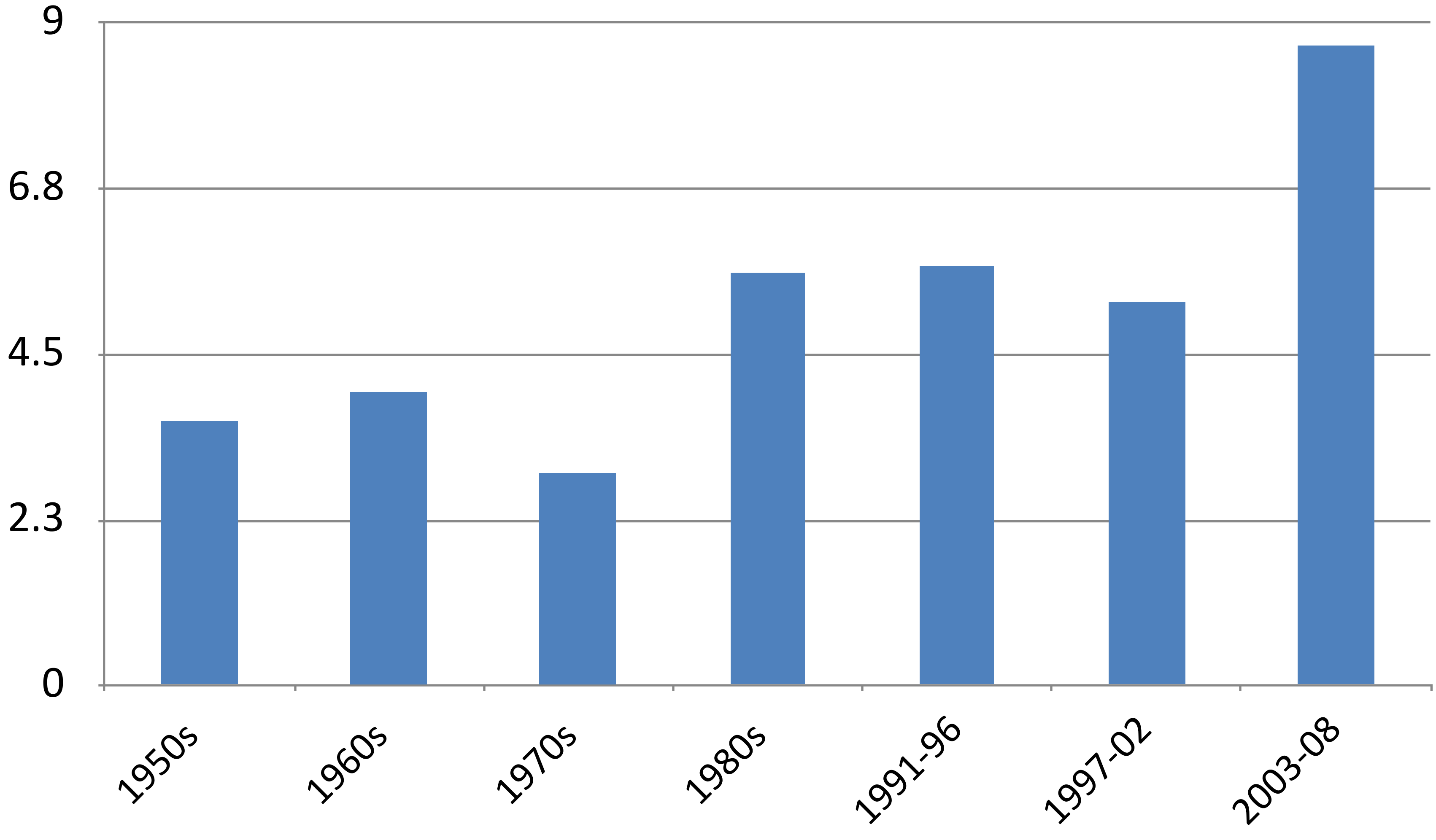
FIGURE 19-6. Aggregate Price and Output Are Determined by the Interaction of Aggregate Supply and Demand

AD and AS curves

- Are analogous to the demand and supply curves that we studied in microeconomics
- But
  - In micro, the demand and supply curves determine quantity and prices of individual commodities
  - In macro, the AD and AS curves determine total output and overall price level

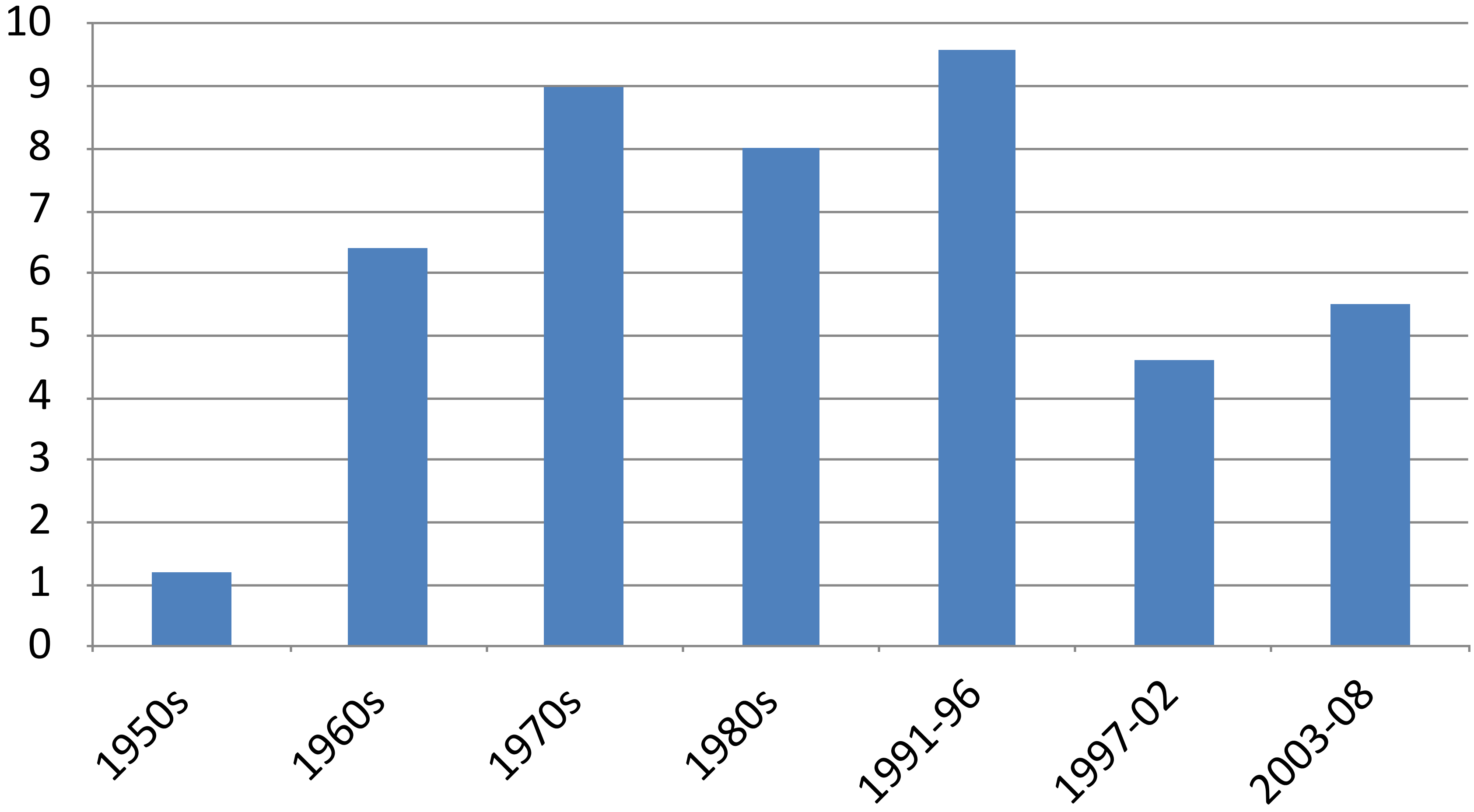
Growth of real GDP: India

Source: Mohan (2008)



Inflation (WPI): India

Source: Mohan (2008)



India’s growth path (1)

Source: Panagariya (2007), Joshi and Little (1994)

- How did India escape prolonged stagnation or decline, unlike the most countries in Africa and Latin America?
- Macroeconomic stability (relatively)
  – Political stability
  – Gradual policy changes
    - Also, “monetary and fiscal conservatism… distinguishes India from most other developing countries”
  – Capacity to implement policies
    - Bureaucracy
    - Institutions such as the Planning Commission and NITI Aayog, RBI, Finance Commission

India’s growth path (2)

Source: Joshi and Little (1994)

- 1970s and early 1980s: oil price shocks had a severe impact on most economies of the world
  - India too very much depends on oil imports, GDP is affected (in particular, manufacturing sector), these shocks also cause inflation and result in BoP issues
  - But, in India, “external shocks are less important than the severe droughts that occur from time to time”
    - This is true mostly for the earlier period (Joshi and Little examine the period, 1960s - 1990s)
  - Also, unlike most countries, India did not borrow heavily in the aftermath of the shocks (countries borrow to fuel investment, and it led to an (temporary) investment boom). Inflation was relatively low, and food and forex reserves were maintained
  - All these meant that the country did not experience a debt crisis or recession in the 1980s
- 1980-82: recessions and external debt problems (in several countries)
  - In India, very high levels of fiscal deficit. Ultimately, a debt crisis followed
- 1990s: Structural adjustment